

Unusual Does Not Mean Impossible

Unit 2 · Topic 2.10 · TODAY'S PROBLEM

Suppose the same 70 percent free-throw shooter takes 100 attempts under comparable conditions and makes 58. Let X be the number of makes if the career-rate model still applies.

Coach: “Anything under 60 makes is impossible for a 70 percent shooter, so 58 proves the career rate no longer applies.”

Analyst: “A result can be unusual without being impossible; compare 58 with the model’s mean and standard deviation.”

The binomial model gives a chance of about 0.0072 for 58 or fewer makes; use this provided value.

Career-rate model

Set	Attempts	Rate	Makes
Tonight	100	0.70	58

FIRST TAKE (5 min, on your sheet)

Whose statement is more defensible, the coach’s or the analyst’s? Name one quantity you would check.

DOK 1 (a) State n and p for this model.

DOK 2 (b) Calculate the binomial mean and standard deviation. Interpret the mean for many sets of 100 shots.

DOK 3 (c) Whose statement is supported? Compare 58 with the mean and standard deviation, and use the given probability to distinguish unusual from impossible. Explain what the coach’s claim could make a reader wrongly believe. Write 3–4 sentences.

Optional review: [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

